

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

Key concept

AN OHMIC CONDUCTOR IS ONE WHOSE RESISTANCE DOES NOT CHANGE

For an ohmic conductor **at a constant temperature**, the current through it is **directly proportional** to the potential difference across it:

$$I \propto V \quad \text{so} \quad R = \frac{V}{I} \text{ stays constant}$$

On an I - V graph that is a **straight line through the origin**. Double the pd, double the current. A fixed resistor at room temperature is the example.

The four words that matter: at a constant temperature. Every ohmic conductor stops being ohmic if you let it get hot.

Non-ohmic: the resistance changes as you use it

The resistance of a **lamp, diode, thermistor** or **LDR** is **not constant** — it changes with the current through the component. Their I - V graphs are **not** straight lines.

The filament lamp — and why it bends. Follow the chain, it is worth three marks on its own:

larger current → filament gets **hotter** → the metal ions **vibrate more** → electrons **collide** with them more often → **resistance increases**

So the current rises *less and less* for each extra volt, and the line bends towards the V axis — the same way whichever direction you drive it.

The diode. Current flows in **one direction only** — forward, once the pd is above roughly 0.6 V. In **reverse** it has a **very high resistance**, so essentially no current flows.

THE TRAP — and it is the one that separates grades here.

Resistance is **always** $R = \frac{V}{I}$ — and on a *curve* you must read **one point** off the graph and divide. Read V across, read I up, divide.

The gradient of an I - V curve is NOT the resistance. On a straight line through the origin the gradient happens to equal $1/R$, which is why people get away with it — but the moment the line bends, the two are different numbers. **Always go back to V/I .**

Worked example

A filament lamp is tested. At $V = 2.0$ V the current is 0.40 A. At $V = 6.0$ V the current is 0.60 A. Show that its resistance increases.

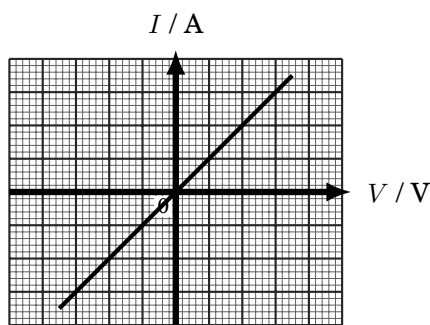
At the first point: $R = \frac{V}{I} = \frac{2.0}{0.40} = 5.0 \, \Omega$

At the second point: $R = \frac{V}{I} = \frac{6.0}{0.60} = 10 \, \Omega$

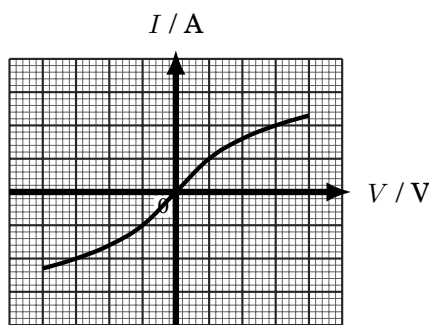
The resistance has **doubled**, because the filament is much hotter at 6 V. Both calculations used a *single point* — neither used a gradient.

The three characteristics

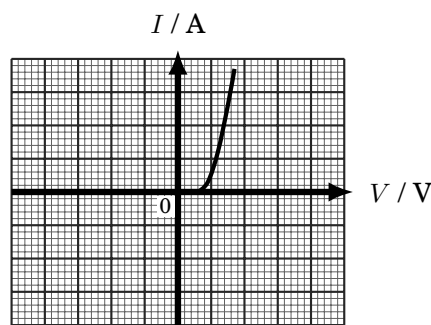
On every graph below: one large square across = 1 V, one large square up = 0.20 A.



Graph A



Graph B



Graph C

1. State what is meant by an **ohmic conductor**. [2]

A conductor whose resistance stays constant (at constant temperature), so the current through it is directly proportional to the pd across it.

2. Name the component that produced each graph. Choose from: **filament lamp**, **diode**, **fixed resistor at constant temperature**. [3]

A: *fixed resistor*

B: *filament lamp*

C: *diode*

3. Which *one* of the three graphs shows a component with **constant resistance**? Explain how the shape tells you that. [2]

Graph A. It is a straight line through the origin, so I is directly proportional to V ; V/I is therefore the same at every point on the line, which means R is constant.

4. Explain, in terms of the filament, **why** graph B bends the way it does. [3]

As the current increases the filament gets hotter. The metal ions in the filament vibrate more, so the electrons collide with them more often, and the resistance increases. So each extra volt produces less extra current and the line bends towards the V axis.

5. Use **graph B** to find the resistance of the lamp at $V = 1.0\text{ V}$ and at $V = 3.0\text{ V}$. Show your working, and state what has happened to the resistance. [4]

At $V = 1.0\text{ V}$, $I = 0.20\text{ A}$: $R = V/I = 1.0/0.20 = 5.0\ \Omega$.

At $V = 3.0\text{ V}$, $I = 0.40\text{ A}$: $R = V/I = 3.0/0.40 = 7.5\ \Omega$.

*The resistance has **increased**, because the filament is hotter at the higher pd. Accept readings within one small square; the required physics is V/I at a point and an increase.*

6. Jamie looks at graph B and says: “The resistance of the lamp is the gradient of the graph.” Explain why Jamie is wrong, and state what he should do instead. [3]

The graph is a curve, so it has no single gradient — and in any case resistance is V/I , not $\Delta I/\Delta V$. On a straight line through the origin the gradient happens to equal $1/R$, which is where the idea comes from, but that only works for a straight line through the origin. He should read one point off the curve and calculate $R = V/I$ at that point.

7. Look at the **left-hand half** of each graph, where the pd has been reversed. [4]

(a) Describe what graph C does when the pd is reversed, and explain it.

Almost no current flows — the line lies flat along the V axis. A diode conducts in one direction only; in the reverse direction it has a very high resistance.

(b) Graph B is symmetrical about the origin. What does that tell you about a filament lamp?

It behaves the same way whichever direction the current flows — it lights and its resistance rises with temperature in both directions. It is non-ohmic, but it is not one-way.

8. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its I – V graph, and explain in one sentence. [2]

it begins to bend over it stays perfectly straight it flattens along the axis

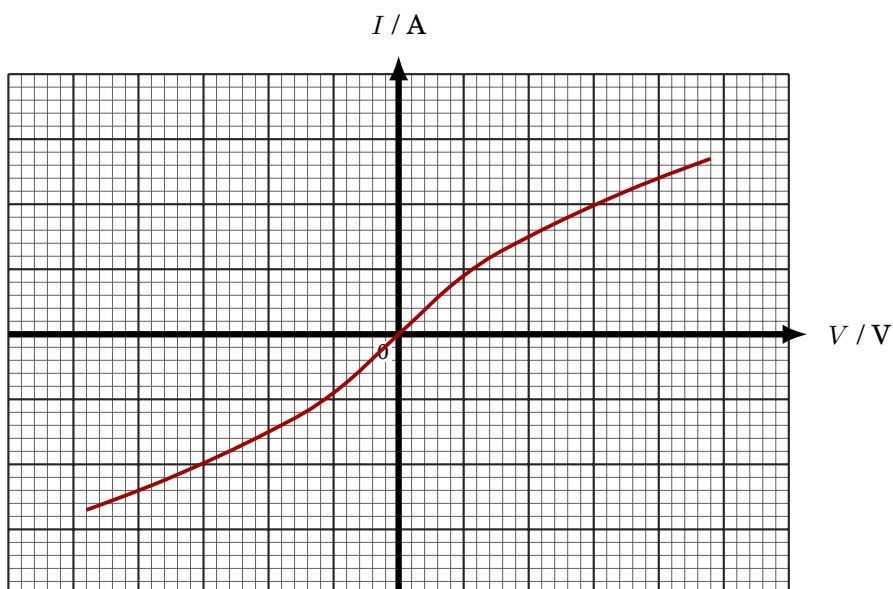
Its resistance is no longer constant, because it is no longer at a constant temperature — so the line starts to curve and the resistor stops behaving ohmically.

9. The resistance of a **thermistor** and of an **LDR** also changes. Complete both sentences. [2]

The resistance of a thermistor _____ as the temperature _____. *decreases ... increases*

The resistance of an LDR _____ as the light intensity _____. *decreases ... increases*

10. Sketch the I – V characteristic you would expect for a **filament lamp**. Use both halves of the axes. [3]



Steepest near the origin, bending towards the V axis on both sides, passing through the origin, symmetrical. One mark each for: through the origin; curving (not straight); both quadrants drawn.